

# Pareto-Optimal Quantized ResNet Is Mostly 4-bit

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## Abstract

Quantization has become a popular technique to compress neural networks and reduce compute cost, but most prior work focuses on studying quantization without changing the network size. Many real-world applications of neural networks have compute cost and memory budgets, which can be traded off with model quality by changing the number of parameters. In this work, we use ResNet as a case study to systematically investigate the effects of quantization on inference compute cost-quality tradeoff curves. Our results suggest that for each bfloat16 ResNet model, there are quantized models with lower cost and higher accuracy; in other words, the bfloat16 compute cost-quality tradeoff curve is Pareto-dominated by the 4-bit and 8-bit curves, with models primarily quantized to 4-bit yielding the best Pareto curve. Furthermore, we achieve state-of-the-art results on ImageNet for 4-bit ResNet-50 with quantization-aware training, obtaining a top-1 eval accuracy of 77.09%. We demonstrate the regularizing effect of quantization by measuring the generalization gap. The quantization method we used is optimized for practicality: It requires little tuning and is designed with hardware capabilities in mind. Our work motivates further research into optimal numeric formats for quantization, as well as the development of machine learning accelerators supporting these formats. As part of this work, we contribute a quantization library written in JAX, which is open-sourced at <https://github.com/google-research/google-research/tree/master/aqt>.

## 1. Introduction

While neural networks have brought tremendous progress to the computer vision field over the last decade,

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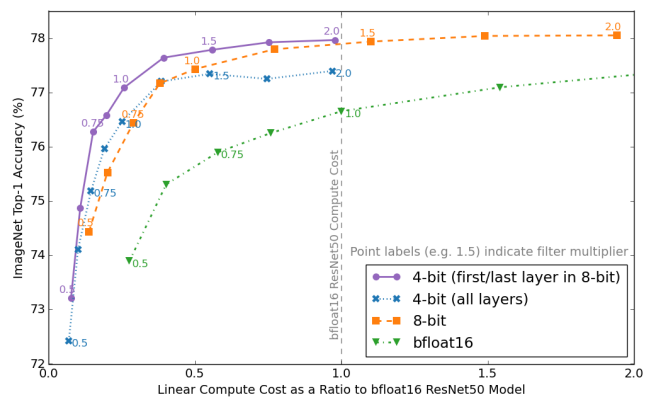


Figure 1. Compute cost-accuracy tradeoff curves, using a linear compute cost model.

they are also often resource-intensive to train and serve. In many real-world computer vision applications, it is therefore of interest to minimize the compute cost (including power consumption/CO<sub>2</sub> output), and the memory footprint, ideally without compromising model quality.

Quantization [12] has become a popular technique to make neural networks, including computer vision models, more efficient. By reducing the number of bits used, quantization helps to compress the model, thus reducing its memory footprint. With support for quantized ops in hardware, e.g. NVIDIA A100 GPUs [22], quantization can also significantly speed up computation and lower power consumption, which can reduce overall compute costs. Quantization directly reduces per-op power-usage, an important factor in both chip design and the cost of ownership. Table 1 shows NVIDIA A100 peak performance in the context of quantization.

Despite its advantages, there could also be downsides to quantizing a model, e.g., a possible reduction in model quality. Figure 2 displays an example showing quantization steps performed on a series of data, e.g. model weights. During quantization, the tensors would be scaled